PCBA Crate Redesign & Shipping Optimization

Schweitzer Engineering Laboratories

PURDUE UNIVERSITY | Edwardson School of Industrial Engineering

SEL

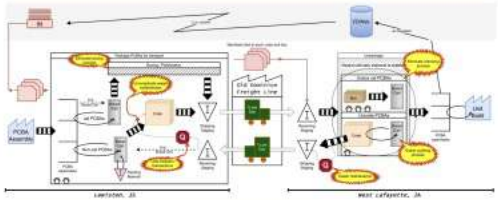
Problem Statement

SEL lacks the ability to optimize its transportation system for delicate PCBAs and reduce transportation costs

Main Issues:

1. Crate design does not support heavy boards
2. Crate maintenance is inefficient and unsafe
3. Lack of standardization in loading & unloading

Current System Model



Analysis & Diagnosis

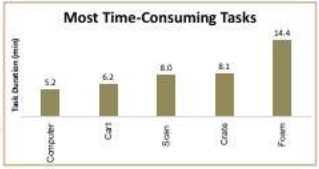
1. Crate Redesign & Durability Upgrades: Unified shipping for all board types, reinforced crate internal fixture and changed pallet type

2. Process & Layout Redesign: Unloading & loading material flow improvements

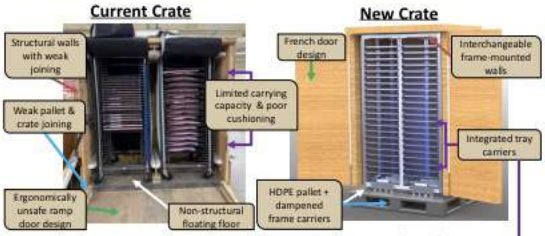
3. Crate Loading Optimization Software: Load-balancing software & website integration

Priority Maintenance Issues (DFMEA)

	Severity
Crate	
• Wall Delamination / Flaring	
• Sheared Door Latches	
• Wall Fork Punctures	
Pallet	
• Total Pallet Disintegration	
• Lead Board Delamination	
• Dislodged Pallet Block	



Crate Redesign & Durability Upgrades



Design Upgrades

- Increased weight capacity
- Supports heavy boards
- ~63% more trays
- In-house assembly
- Built-in ESD-safe packaging
- Reinforced, rigid, dampened joints
- Modular parts for easy maintenance
- French door for easy access

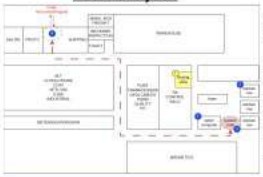
New Self-Locking Tray Carrier

Scan for Showcase & More

Process & Layout Redesign

[redacted] Loading Process

Future Layout



Loading Material Flow

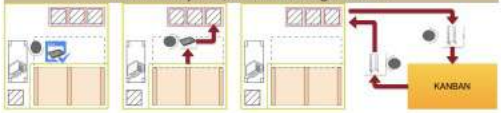
i. Receive orders list CSV, use website for loading instructions

ii. Receive PCBs from Kanban

iii. Move crate to staging/loading area

iv. Return loaded crate to shipping

West Lafayette, IN Unloading Process



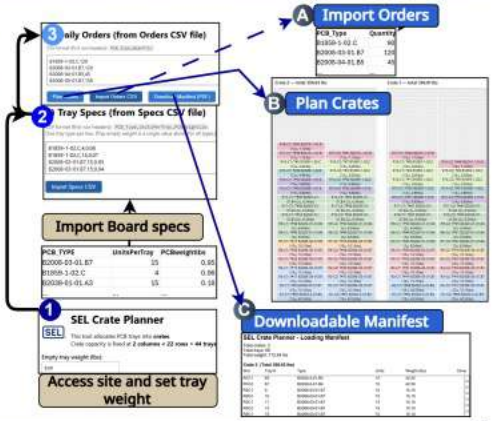
Future Unloading Material Flow

i. Unlock tray cartridge; tilt tray; inspect & audit PCBAs

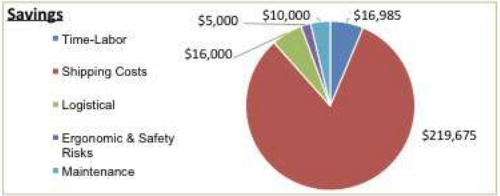
ii. Transfer tray to rolling bread cart; repeat until all trays are unloaded

iii. Move full bread carts to Kanban area

Crate Loading Optimization Software



Impact & Recommendations



Impact

1	Major process eliminated
3	Ergonomic risks addressed
50%	Unloading time reduction
500	Labor hours saved per year
63%	Increase to PCBA capacity
$265K+	Total Yearly Savings

Conclusion

Our proposals will:

- **Enhance safety** during maintenance & loading
- **Eliminate parallel shipping process** to cut cost & complexity
- **Automate load-balancing** & improve tracking accuracy
- **Simplify operations** across loading, packaging & material flow
- **Improve crate cost density**

Team Members: Kevin Bao, [redacted] | **Client Mentors:** Ryan Peck & Scott Kemp